

Math 122: Exam 1 - Practice 1

Student Name: _____

Instructions:

- Show all work: Full credit requires complete, correct steps and proper math notation.
 - Points: Each question is worth 10 points.
 - Formula Sheet: Permitted (formulas only; no worked examples).
 - Calculators: Basic, non-graphing calculators allowed.
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Problem 1

Find the exact arc length of the curve

$$y = 3x^{3/2} - 1$$

over the interval from $x = 0$ to $x = 1$.

Problem 2

Find the limit:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$$

Problem 3

Find the limit:

$$\lim_{x \rightarrow +\infty} x e^{-x}$$

Problem 4

Find the limit:

$$\lim_{x \rightarrow 0} (e^x + x)^{1/x}$$

Problem 5

Find $\frac{dy}{dx}$:

$$y = \sin^{-1}(1/x)$$

Problem 6

Find $\frac{dy}{dx}$:

$$y = e^x \sec^{-1} x$$

Problem 7

Evaluate the integral:

$$\int \frac{dx}{\sqrt{1-4x^2}}$$

Problem 8

Evaluate the integral:

$$\int \frac{\sec^2 x \, dx}{\sqrt{1-\tan^2 x}}$$

Problem 9

Evaluate the definite integral:

$$\int_1^{\sqrt{3}} \frac{\sqrt{\tan^{-1} x}}{1+x^2} \, dx$$

Problem 10

Evaluate the integral:

$$\int \frac{\sin \theta}{\cos^2 \theta + 1} \, d\theta$$

— END OF EXAMINATION —